

Muhammad Ahsan

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Ph.D. Candidate in Computer Science with over 10 years of research experience in cybersecurity, IoT security, machine learning, and cyber-physical systems. My work focuses on securing intelligent and connected physical infrastructures through side-channel analysis and data-driven methods. Committed to integrating research and teaching to advance cybersecurity education, digital forensics, and resilient cyber-physical systems.

Education

Virginia Commonwealth University (VCU)

PhD in Computer Science (CGPA: 4.00)

Richmond, Virginia USA

Aug 2021 - July 2026 (Expected)

University of Engineering and Technology (UET)

Master of Electrical Engineering (CGPA: 3.75)

Lahore, Pakistan

Jan 2015 - July 2018

University of Engineering and Technology (UET)

B.Sc. Electrical Engineering (CGPA: 3.70)

Lahore, Pakistan

Aug 2010 - June 2014

Research Interests

Cybersecurity; Cyber-Physical Systems Security; Secure Additive Manufacturing; Side-Channel Analysis; Artificial Intelligence and Machine Learning for Security; IoT and Embedded Systems Security; Digital Forensics and Malware Analysis.

Funding and Proposals

- "Cybersecurity Initiative in Bioprinting", Virginia Commonwealth Cyber Initiative (CCI), Total Funding: \$100,000 (all direct cost), Project Duration: 1 year (Sept 2022 - Dec 2023)
(Made significant contributions in writing the technical part of the proposal)
- "Securing the Layered Build: Cyber-Physical Defense in Additive Manufacturing Systems of Metal Fused Filament Fabrication", Virginia Commonwealth Cyber Initiative (CCI), Total Funding: \$100,000 (all direct cost), Project Duration: 1 year (2025 - 2026)
(Made significant contributions in writing the technical part of the proposal)

Awards and Achievements

- **Student Travel Awards:** ACSAC 2023 Student Conferenceship, NSF travel funding (2024, 2025, 2026) to attend HOST conference.
- **Best Defender Award**, U.S. Cyber Command Cyber Recon 2025.
- **Best Guardian Award**, U.S. Cyber Command Cyber Recon 2024.
- **Commander's Award for Outstanding Research**, U.S. Cyber Command Cyber Recon 2024.
- **Four-Star Challenge Coin Recognition** presented by General Timothy D. Haugh, Commander, United States Cyber Command and Director, National Security Agency, for research contributions.
- **People's Choice Award**, MegaHack@VCU, 2024, Richmond, Virginia.
- **Dean's Honor Certificate** (awarded for five semesters of B.Sc.), UET Lahore, Pakistan.
- **B.Sc. Electrical Engineering with Honors**, UET Lahore, Pakistan.

Teaching and Mentoring Experience

- **Instructor, PLC Short Courses**, University of Engineering and Technology (UET) Lahore. Designed and delivered short courses on Programmable Logic Controllers (PLCs), covering industrial automation concepts and hands-on laboratory training.
- **Graduate Teaching Assistant, CMSC 355**, Virginia Commonwealth University (VCU). Supported instruction in software engineering concepts through laboratory assistance, student mentoring, assignment evaluation, and technical guidance.
- **Invited Speaker, Seminar Series**, Virginia Commonwealth University (VCU). Delivered invited research seminars on cybersecurity and cyber-physical systems, engaging graduate and undergraduate audiences.
- **Undergraduate Research Mentorship** Mentored undergraduate students on research projects involving cyber-physical systems, IoT platforms, and cybersecurity experimentation.
- **Curriculum Development Contributor**, sequa gmbH – GIZ Collaboration Contributed to the design and development of an Energy Efficiency Advisory course, creating training materials and technical curriculum focused on industrial energy efficiency and sustainable engineering practices.

Conference/Journal Publications

1. **Muhammad Ahsan**, Johanna Tsala Ebode, Nastassja Lewinski, Irfan Ahmed, "BioShield: A Power Side-Channel–Based Monitoring Framework for Ensuring 3D Bioprinting Process Integrity", Network and Distributed System Security (NDSS) Symposium, **Under review**.
(CORE Ranking: A)
2. Muhammad Haris Rais, **Muhammad Ahsan**, Irfan Ahmed, "PrintSafe - A near real-time anomaly detection framework for fused filament fabrication printing using printing environment estimation", Journal of Manufacturing Processes, **Under review**.
(Q1 Journal with an impact factor of 6.03 in 2023)
3. **Muhammad Ahsan**, Eunice Pak, Kate Jackson, Muhammad Haris Rais, Barry Najarro-Blancas, Nastassja Lewinski, Irfan Ahmed, "BioSaFe: Bioprinting Security Framework for Detecting Sabotage Attacks on Printability and Cell Viability", In the 40th IEEE Annual Computer Security Applications Conference (ACSAC), Dec 2024, Hawaii.
(Acceptance rate (21.5%): 83 out of 381 submissions)
(CORE Ranking: A)
(Included in the VA Commonwealth Cyber Initiative's 2025 CCI Research Showcase, a curated collection of significant, influential peer-reviewed research papers published in the last two years.)
4. **Muhammad Ahsan**, Irfan Ahmed, "WattShield: A Power Side-Channel Framework for Detecting Malicious Firmware in Fused Filament Fabrication", In the 18th IEEE International Symposium on Hardware Oriented Security and Trust (HOST), May 2025, San Jose, CA.
(Acceptance rate (31.57%): 42 out of 133 submissions)
5. **Muhammad Ahsan**, Barry Najarro-Blancas, Johanna Tsala Ebode, Nastassja Lewinski, Irfan Ahmed, "3D Bioprinter Firmware Attacks: Categorization, Implementation, and Impacts", In the 18th IEEE International Symposium on Hardware Oriented Security and Trust (HOST), May 2025, CA.
(Acceptance rate (31.57%): 42 out of 133 submissions)
(Semi-Finalists for the Best Paper Award - among the top 6 papers)
6. Muhammad Haris Rais, **Muhammad Ahsan**, Irfan Ahmed, "SOK: 3D Printer Firmware Attacks on Fused Filament Fabrication." In the 18th USENIX WOOT Conference on Offensive Technologies (WOOT), August 2024, Philadelphia, PA.
(Acceptance rate (35%): 18 out of 51 submissions)
7. **Muhammad Ahsan**, Muhammad Haris Rais, and Irfan Ahmed, "SOK: Side Channel Monitoring

for Additive Manufacturing-Bridging Cybersecurity and Quality Assurance Communities." In the 8th IEEE European Symposium on Security and Privacy (**EuroS&P**), July 2023, Netherlands.

(Acceptance rate (35%): 63 out of 180 submissions)

(CORE Ranking: A)

(Included in the VA Commonwealth Cyber Initiative's 2024 CCI Research Showcase, a curated collection of significant, influential peer-reviewed research papers published in the last two years.)

8. Muhammad Haris Rais, **Muhammad Ahsan**, Irfan Ahmed, "Sabotaging Material Extrusion-Based 3D Printed Parts through Low-Magnitude Kinetic Manipulation Attacks." In ACM Transactions on Cyber-Physical Systems (TCPS), Vol. 9, No. 1, pp. 1-26, January 2025.
(Q1 Journal with an impact factor of 2.3 in 2023)
9. Bilal Imran, **Muhammad Ahsan**, Ali Hammad Akbar, and Ghalib Asadullah Shah. "D4gw: Dtls for Gateway Multiplexed Application to Secure MQTT (SN)-Based Pub/Sub Architecture.", In Elsevier **Internet of Things Journal**. 2024.
(Q1 Journal with an impact factor of 7.6 in 2024)
10. Muhammad Haris Rais, **Muhammad Ahsan**, and Irfan Ahmed. "FRoMEPP: Digital Forensic Readiness Framework for Material Extrusion based 3D Printing Process", In 10th Annual Digital Forensics Research Conference Europe (DFRWS EU'23), March 2023, Bonn, Germany
- Published by Forensic Science International (FSI): Digital Investigation Journal, Elsevier.
(Acceptance rate (23%): 16 out of 70 submissions)
11. **Muhammad Ahsan**, and Muhammad Ali. "LsStk: Lightweight solution to preventing Stack from buffer overflow vulnerability", 2023 IEEE 17th International Conference on Open Source Systems and Technologies (ICOSST), 2023.
12. Muhammad Rais, **Muhammad Ahsan**, Vaibhav Sharma, Radhika Barua, Robert Prins, and Irfan Ahmed, "Low-Magnitude Infill Structure Manipulation Attacks on Fused Filament Fabrication Printers", In the 16th IFIP International Conference on Critical Infrastructure Protection (ICCIP), March 2022, Arlington, Virginia.
13. **Muhammad Ahsan**, Bilal Afzal, Bilal Imran, Asim Tanwir, Ali Hammad Akbar, and Galib. A. Shah. "OneM2M Architecture Based Secure MQTT Binding in Mbed OS", IEEE **Euro S&P** Workshop on Software Security for Internet of Things. 2019.
14. Bilal Imran, Bilal Afzal, Dr. Ali Hammad Akbar, **Muhammad Ahsan**, Dr. Ghalib A. Shah. "MISA: Minimalist Implementation of oneM2M Security Architecture for Constrained IoT Devices", IEEE **Globecom** 2019.
(Acceptance rate (40%): 69 out of 172 submissions)
(CORE Ranking: B)
15. **Muhammad Ahsan**, Naseer Ahmad, Waqas Badar. "Simulation of Solar angles for maximizing Efficiency of Solar Thermal Collectors", 3rd Inter. Conference on Energy Conservation and Efficiency (ICECE), 2019.

Peer reviewed Tutorials/Hardware-Demos/Posters

1. "When Code Meets Cells: Securing 3D-Bioprinting Systems against Adversarial Manipulations", Poster **CCI Symposium 2026**
2. "Supply Chain Resilience in Industrial Control Systems", HW tutorial **HOST Conference 2026**
3. "When Code Meets Cells: Securing 3D Bioprinting Systems Against Adversarial Manipulation", HW Demo **HOST Conference 2026**
4. "3D Bioprinter Firmware Attacks and their Implications" Poster **CCI Symposium 2025**
5. "Firmware Attacks on Inkredible+ 3D Bioprinter" HW Demo **HOST Conference 2025**
6. "A Framework for detecting Sabotage attacks on 3D Bioprinters using Side Channel Monitoring" Poster **CCI Symposium 2024**
7. "BioSaFe: Bioprinting security framework for in-situ process monitoring", HW Demo **HOST**

Conference 2024

8. "The Troubled Elevator: Forensic Investigation of a Bank's Elevator Malfunctioning" **DFRWS 2023**
Challenge on Industrial Control System Forensics
9. "Bioprinting Security Framework for detecting Sabotage attacks using in situ Process Monitoring",
Poster **ACSAC Conference 2023**

Industry/Academia Research Experience

Academic Research.....

Security and Forensics Engineering (SAFE) LAB

VCU, Richmond

Position: Graduate Assistant

Aug 2021 - Current

Project Title: Side Channels for Securing Cyber-Physical Systems (CPS).

Description: My research advances security for additive manufacturing by leveraging side-channel sensing and machine learning to monitor and protect the end-to-end printing process. I study both attack surfaces and defense mechanisms in polymer 3D printing and bioprinting, with the goal of establishing trustworthy and resilient cyber-physical manufacturing systems.

Applied/Industry Research.....

Al-Khawarizmi Institute of Computer Science (KICS)

UET, Lahore

Position: Sr. Research Officer

Oct 2014 - July 2021

- **Lightweight Security Framework for IoT:** Designed a lightweight security framework addressing interoperability and security challenges in IoT ecosystems aligned with the oneM2M standard, focusing on secure device communication and scalable deployment.
- **Smart Home Cyber-Physical Platform:** Designed and implemented a customized smart home testbed, including development of an embedded gateway enabling secure integration and control of smart plugs and switchboards for experimentation with IoT system architectures.
- **LoRaWAN-based Smart Irrigation System:** Developed a long-range IoT sensing framework using LoRaWAN for real-time acquisition, processing, and mobile visualization of soil moisture data to support data-driven precision agriculture.
- **Security Analysis of IoT/SCADA Firewalls:** Designed and evaluated firewall rule sets for industrial IoT and SCADA protocols through vulnerability-driven testing and adversarial network traffic generation.
- **Industrial Automation Systems:** Developed automation solutions for agricultural and infrastructure applications, including paddy dryer control in rice mills and automated monitoring of large-scale water storage systems.
- **Solar Thermal Tracking Systems:** Contributed to the design and implementation of control algorithms for solar tracking in industrial heliostats and parabolic trough collectors to improve energy harvesting efficiency.
- **IEC-Compliant Motor Test Bench:** Designed and developed an industrial motor test platform compliant with IEC 60034-2-1 standards for evaluation of motor performance and efficiency.

Major Subjects

Memory and Malware Forensics
Network Security
Advanced Operating Systems

High-Performance Distributed Systems
Design and Analysis of Algorithms
Machine Learning

Academic Service

- **Conference Organization and Management:** Contributed to conference organization and management activities for IEEE ICOSST (2016–2019) and IEEE ICECE (2017, 2018), supporting technical program coordination, reviewing process, and event operations.
- **Journal Reviewer:** Served as a peer reviewer for leading journals including IEEE Internet of Things Journal, ACM Transactions on Cyber-Physical Systems (TCPS), Forensic Science International: Digital Investigation, and Computers & Security.

Professional References

1. **Irfan Ahmed, Professor**
 - Department of Computer Science, Virginia Commonwealth University
 - Email: iahmed3@vcu.edu
2. **Nastassja Lewinski, Associate Professor**
 - Department of Chemical and Life Science Engineering, Virginia Commonwealth University
 - Email: nalewinski@vcu.edu
3. **Radhika Barua, Assistant Professor**
 - Department of Mechanical and Nuclear Engineering, Virginia Commonwealth University
 - Email: rbarua@vcu.edu